Time–Frequency Methods in Index Tracking

June 22, 2026

1 Introduction

Portfolio optimisation and index replication have long relied on the statistical relationship between asset prices, yet the financial econometrics literature has never reached a settled consensus on which measure of dependence is most appropriate for these tasks. Two candidates have dominated the debate: correlation, which quantifies the co-movement of asset returns over a fixed sample window, and cointegration, which identifies the existence of a common stochastic trend in the price levels of two or more integrated series. Though both concepts are grounded in the same probabilistic foundations, they capture fundamentally different aspects of dependence, and the choice between them has non-trivial consequences for portfolio construction, risk management, and trading strategy design.

This paper argues that the debate has been incompletely framed. Flip flops and mountaineering boots are both footwear. Neither is wrong. The difficulty arises when one attempts to declare a universal winner across all terrains: the question of which is better has no answer until the terrain is specified. The correlation versus cointegration literature has largely proceeded as though the terrain were fixed, seeking to settle which approach is superior without first asking what the investor's horizon actually requires.

The resolution lies in frequency analysis. Correlation and cointegration are not competing methods but limiting cases of a single family of tracking criteria, each targeting a different part of the frequency spectrum of the portfolio-benchmark spread. This becomes precise through the frequency response of the differencing operator: working in returns concentrates the tracking objective on fluctuations at timescales of days, while working in price levels concentrates it on fluctuations at timescales of years. Neither is universally better. Each is well suited to a specific investor horizon and poorly suited to all others.

Fractional integration provides a natural interpolation between these two extremes, with each value of the fractional differencing parameter d corresponding to a concrete intermediate horizon. But the fractional family is itself limited: it can only produce monotone frequency responses, and real investor preferences are not always monotone. A general frequency response relaxes this restriction entirely, allowing the tracking criterion to be shaped to any horizon profile. The result is a framework in which investor time horizon preference is a design parameter, expressible quantitatively and continuously, rather than a binary choice between two inherited methodologies. The present work proposes such a framework, grounded in time-frequency representations of the price process. This is extended to linear risk-coherent optimisation programmes [Artzner et al., 1999] for both index tracking and enhanced index tracking, and portfolio sparsity is enforced through the ℓ_0-norm approximation procedure of Benidis et al. [2018].

1.1 The Correlation Paradigm and Its Limitations

The correlation-based paradigm has its roots in the mean-variance framework of Markowitz [1959] and was formalised for index tracking by Roll [1992], whose tracking error variance minimisation model remains the practitioner standard. The appeal of correlation is its familiarity and computational tractability: given a sample of returns, a covariance matrix can be estimated and an optimal portfolio derived in closed form. The limitation, however, is equally well understood. Correlation is computed from return series, which are stationary by construction, and therefore captures only short-run co-movements. Alexander [1999] observed that correlation is intrinsically a short-run measure and that tracking portfolios built on it carry no guarantee that the price spread between the replica and the benchmark will remain bounded over time. For daily data, this short-run character is pronounced: the tracking objective is primarily sensitive to fluctuations at horizons of days to weeks, and is essentially blind to slower-moving divergences between the portfolio and the benchmark. This is not an empirical accident but a mathematical consequence of working with return series, which are first differences of prices: differencing acts as a high-pass filter, suppressing low-frequency trends and retaining only rapid fluctuations [Chaudhuri and Lo, 2016, Engle, 1974]. Section **??** makes this precise and quantifies the effective horizon of correlation-based tracking. In the absence of a mean-reversion mechanism, the tracking error may behave as a random walk, drifting arbitrarily far from zero and necessitating frequent and costly rebalancing. This observation, which has since been reproduced in numerous empirical settings, motivates the cointegration alternative.

1.2 The Cointegration Framework

The cointegration framework, introduced by Granger [1981] and Engle and Granger [1987] and later extended by Johansen [1988, 1991], addresses the deficiency of correlation by working directly with price levels. Two or more $I(1)$ series are said to be cointegrated if a stationary linear combination of them exists. In the index tracking context this is a natural property: because a market index is itself a linear combination of its constituent prices, there must in principle exist a portfolio whose value tracks the index with a stationary, mean-reverting error. Alexander [1999] formalised this insight and showed that a cointegration-optimal portfolio guarantees stationarity of the tracking error by construction, ties the portfolio to its benchmark in the long run, and produces more stable portfolio weights because they are estimated from price levels over a long calibration window. For daily data, calibration windows of three to five years are standard, and the method is largely insensitive to fluctuations at horizons shorter than a year. This long-run character is the structural counterpart to correlation's short-run sensitivity: working in price levels, which are the cumulative sum of returns, acts as a low-pass filter, suppressing rapid fluctuations and retaining only slow-moving common trends [Chaudhuri and Lo, 2016]. Section **??** quantifies this precisely. A direct consequence is that cointegration-based portfolios require less frequent rebalancing than their correlation-based counterparts, not as a practical convenience but because the objective itself is insensitive to the short-run deviations that would otherwise trigger rebalancing.

These theoretical properties were subsequently developed into explicit trading strategies by Alexander et al. [2002] and Alexander and Dimitriu [2002, 2005a,b], who demonstrated that cointegration-based long-short equity strategies can achieve low volatility, near market-neutrality, and favourable Sharpe ratios across multiple market regimes. The error correction model that underpins cointegration reveals causal flows, sometimes called Granger causalities, that must exist between series sharing a common stochastic trend [Granger, 1988], and this dynamic structure is itself a source of useful information for portfolio management and risk control.

1.3 An Open and Inconclusive Empirical Debate

1.3.1 Evidence from the Literature

The empirical evidence, however, is considerably less clear-cut than the theoretical arguments would suggest. Alexander and Dimitriu [2005a], in what remains the most rigorous direct comparison, found that for simple index tracking using Dow Jones Industrial Average constituents the out-of-sample performance of cointegration-optimal portfolios was broadly similar to that of tracking error variance minimisation, with neither method exhibiting a decisive advantage once the number of stocks was sufficient. Grobys [2010], applying the same comparison to the Swedish OMX market over a ten-year period, reached the opposite conclusion, finding that cointegration-based portfolios clearly dominated on both Sharpe and Treynor ratios, with the best portfolio outperforming the index by a substantial margin at lower volatility. Sant'Anna et al. [2016], who made portfolio selection endogenous to the cointegration optimisation through large-scale simulation across Brazilian and United States data, found again that neither method consistently dominates: cointegration delivered better tracking error for Brazilian equities while correlation was superior on turnover and volatility, and results for the United States market were mixed.

In a different application of the same two building blocks, Miao [2014] proposed a two-stage procedure in which correlation filters candidate pairs and cointegration provides the final selection criterion for high-frequency statistical arbitrage, reporting strong out-of-sample performance on energy sector stocks. The combination of the two stages is telling: it suggests that each measure captures something the other misses, and that a unified treatment would be preferable to sequential filtering. Arsov [2023], examining small South-East European markets, found that although Johansen tests initially indicated cointegration, the result could not be validated diagnostically, and the analysis reverted to vector autoregressive and GARCH frameworks. The broader literature on emerging market integration and volatility spillovers similarly finds that the presence or absence of cointegration is sensitive to market structure, data frequency, and sample length, with no universal verdict.

Beyond the index tracking literature, similar ambiguity appears in studies of international market linkages. A recurring finding is that developed markets transmit both return and volatility shocks to smaller markets through mechanisms that operate at multiple time scales simultaneously [Arsov, 2023, Alexander and Dimitriu, 2005b]. These multi-scale transmission patterns cannot be adequately characterised by either correlation, which aggregates across all frequencies, or cointegration, which focuses exclusively on the lowest.

1.3.2 A Question of Investor Horizon

The inconclusiveness of this literature is less surprising once the horizon argument is taken seriously. The studies above largely evaluate both methods against the same short-run tracking error metrics, regardless of the horizon the method was designed for. This is precisely where the terrain matters. Asking which method is better without specifying the investor's horizon is not a well-posed question.

Consider two investors. The first is a portfolio manager running a short-term equity overlay, rebalancing weekly to maintain index exposure for hedging purposes. Daily and weekly deviations from the benchmark directly affect the hedge's effectiveness and need to be kept small. Multi-year drift is irrelevant: positions are rarely held beyond a few months. For this investor, correlation-based tracking is the natural choice. The tracking objective is sensitive to exactly the fluctuations that matter, and frequent rebalancing is not a cost to be minimised but a feature of the strategy.

The second is an asset manager running the equity sleeve of a defined benefit pension fund, matching assets to liabilities that extend twenty years into the future. Whether the portfolio was 0.3% away from the benchmark on any given Tuesday is of no consequence. What matters is that when liabilities fall due, the portfolio value is close to the benchmark value. Rebalancing costs compound over decades, so stability of weights is a primary concern. For this investor, cointegration-based tracking is the natural choice. The long-run adherence it guarantees is exactly what the liability-matching mandate requires, and the lower turnover is a direct benefit.

Neither investor is wrong about their method. They are simply operating at different points on the frequency spectrum of the tracking error. The empirical debate, by evaluating both methods on a common metric without regard to horizon, has been asking which of the two serves both investors simultaneously, and correctly finding that neither does so consistently.

This reframing points naturally toward an intermediate territory. A quarterly-reporting fund, a retail investor with a five-year savings horizon, or a risk-parity manager rebalancing monthly all sit between these two extremes and are served by neither method in its pure form. The theory of fractional integration, introduced by Granger and Joyeux [1980] and Hosking [1981], provides a continuous interpolation. By allowing the order of differencing d to take non-integer values between zero and one, it produces tracking criteria whose effective horizon varies continuously from days to years as d decreases from one toward zero. Correlation corresponds to $d = 1$; cointegration corresponds to $d \approx 0$; and every investment horizon in between corresponds to some intermediate value. Section 2.1 develops this connection formally, and Section **??** makes the horizon mapping precise.

1.4 A False Dichotomy

What emerges from this body of work is not a resolved debate but an incompletely specified one. The studies reviewed above have largely asked which method is better, without first specifying better for whom and over what horizon. As the two investor examples in the previous section illustrate, this question has no horizon-independent answer: correlation is the correct tool for the equity overlay manager hedging weekly exposures, and cointegration is the correct tool for the pension fund matching twenty-year liabilities. Evaluated on the same short-run tracking error metric, cointegration will appear inferior. Evaluated on long-run price-level adherence, correlation will appear inferior. Neither conclusion is informative about the methods themselves; both reflect the mismatch between the evaluation criterion and the method's intended horizon. This is the primary source of the inconclusiveness documented in Section 1.3.

Beyond this misspecification, the discussion has also proceeded in binary terms: a portfolio is either built on correlation or on cointegration, the tracking error is either stationary or it is not, and the analyst either differences the data or regresses in levels. Consider a third investor: a target-date retirement fund manager with a ten-year horizon, rebalancing quarterly and reporting annual performance to clients. Daily noise is irrelevant to this investor, but so is the exclusive focus on decade-scale price-level adherence that cointegration enforces. This investor sits squarely between the two extremes, and the binary framing offers no natural home for them. The dichotomy is therefore a second and separate limitation, compounding the primary one.

Correlation and cointegration are not competing definitions of dependence so much as limiting cases of a richer structure. Correlation, by operating on returns, is sensitive to high-frequency co-movements while being blind to low-frequency common trends. Cointegration, by identifying stochastic trends in price levels, is precisely a statement about the very lowest frequencies of the data generating process: the near-zero portion of the spectrum where power accumulates without bound in an integrated series. The two approaches are therefore not in opposition; they are each sampling a different part of the spectral content of the multivariate price process.

This observation connects naturally to the theory of fractional integration. A series that is $I(0)$ is stationary and its spectral density is bounded and flat at the origin. A series that is $I(1)$ has a spectral density that diverges at zero frequency, reflecting the accumulation of permanent shocks. Fractional integration, in the sense of Granger and Joyeux [1980] and Hosking [1981], allows the order of integration to take non-integer values between zero and one, producing series whose spectral density diverges at the origin at a rate that interpolates continuously between the stationary and the unit-root cases. From this perspective, the correlation-versus-cointegration debate is a debate about whether the tracking error lives at $d = 0$ or at some positive fractional value approaching unity, and the practitioner's choice of optimisation method implicitly encodes an assumption about which frequency bands are relevant to the investment horizon in question. Section 2.1 makes this mapping precise, providing a concrete correspondence between values of d and investment horizons in trading days.

The fractional differencing family, however, imposes a fundamental restriction: its frequency responses are necessarily monotone. For any value of d, the resulting filter either up-weights low frequencies or high frequencies, and cannot selectively target an intermediate band. An investor whose mandate is defined at a specific horizon, such as a quarterly-reporting fund that is indifferent to both daily noise and multi-year drift, cannot express this preference through any choice of d. The generalisation to arbitrary frequency responses, developed in Section 3.4, removes this restriction entirely.

1.5 Structure of the Paper

The remainder of this paper is organised as follows. Section 2 establishes the mathematical foundations required for the analysis. It introduces fractional differentiation and its relationship to long-memory processes (Section 2.1), presents the discrete Fourier transform and the frequency response as analytical tools (Section 2.2), and motivates the move to time-frequency representations (Section 2.3). It then sets out the index tracking and enhanced index tracking optimisation problems (Section 2.4), reviews coherent risk measures and their linear programme formulations (Section 2.5), and introduces sparse index tracking (Section 2.6).

Section 3 develops the core argument quantitatively, mirroring the narrative of Section 1 with precise derivations. It derives the frequency responses of the differencing and integration operators and connects them to concrete investor horizons via the standard signal processing half-power cutoff (Sections 3.1 and 3.2). It then derives the fractional differencing family and maps each value of d to a trading-day horizon (Section 3.3). It demonstrates that passband representations allow more precise horizon targeting than the monotone fractional family permits (Section 3.4), and argues that the non-stationarity of financial returns necessitates time-frequency rather than purely Fourier-based analysis (Section 3.5).

2 Foundations

2.1 Fractional Differentiation

The distinction between the correlation and cointegration paradigms rests on whether one works with returns or with prices, that is, on whether the price series is differenced once or not at all. This is a binary choice between two integer orders of differencing. Fractional differentiation generalises the differencing operation to non-integer orders, and in doing so replaces the binary choice with a continuous one.

It is convenient to express differencing through the lag operator L, defined by its action of shifting a series one step into the past,

$$Lx_t = x_{t-1}, \qquad L^k x_t = x_{t-k}. \tag{1}$$

The first difference of a series, the change from one period to the next, is then

$$\Delta x_t = x_t - x_{t-1} = (1-L)x_t, \tag{2}$$

so that the operation of differencing is represented by the operator $(1-L)$. Differencing d times corresponds to applying this operator d times, that is, to raising it to the power d.

Because $(1-L)^d$ is a binomial raised to a power, it expands according to the binomial theorem,

$$(1-L)^d = \sum_{k=0}^{d} \binom{d}{k} (-L)^k, \tag{3}$$

where the coefficients $\binom{d}{k}$ are the familiar binomial coefficients. For $d = 2$, for example, the expansion yields $(1-L)^2 = 1 - 2L + L^2$, so that $(1-L)^2 x_t = x_t - 2x_{t-1} + x_{t-2}$. For any positive integer d the sum terminates at $k = d$, since the binomial coefficients vanish beyond that point, and differencing therefore involves only a finite number of past observations.

The binomial coefficient is conventionally defined through factorials,

$$\binom{d}{k} = \frac{d!}{k!\,(d-k)!}. \tag{4}$$

This definition restricts d to non-negative integers, since the factorial is defined only for such values. To admit non-integer orders of differencing, a generalisation of the factorial to arbitrary real arguments is required.

Such a generalisation is provided by the gamma function, introduced by **?**, which extends the factorial to the real and complex numbers while preserving its defining property,

$$\Gamma(n+1) = n! \quad \text{for non-negative integers } n, \tag{5}$$

and remaining well defined for non-integer arguments. Substituting the gamma function for the factorials in the binomial coefficient gives

$$\binom{d}{k} = \frac{\Gamma(d+1)}{\Gamma(k+1)\,\Gamma(d-k+1)}, \tag{6}$$

an expression that coincides with the ordinary binomial coefficient when d is an integer but remains meaningful for any real d.

With the binomial coefficient defined for arbitrary real orders, the difference operator extends directly to non-integer d. Following Granger and Joyeux [1980] and Hosking [1981], the fractional difference operator of order d is defined by the series

$$(1-L)^d = \sum_{k=0}^{\infty} \binom{d}{k} (-L)^k = \sum_{k=0}^{\infty} \frac{\Gamma(d+1)}{\Gamma(k+1)\,\Gamma(d-k+1)} (-1)^k L^k. \tag{7}$$

Unlike the integer case, the series no longer terminates: for non-integer d the coefficients are never exactly zero, and the operator depends on the entire past of the series. Applied to a series, fractional differentiation is thus a weighted sum of all preceding values,

$$(1-L)^d x_t = x_t - d\,x_{t-1} + \frac{d(d-1)}{2}\,x_{t-2} - \cdots, \tag{8}$$

with weights determined by the generalised binomial coefficients.

The dependence on the entire history raises a question of practical computability, since only a finite sample is ever available. The weights, however, decay as k increases, so the operator can be approximated to arbitrary accuracy by truncating the series at a finite number of lags.

Figure **??** illustrates the effect of the operator on the S&P 500 price series for several orders of differencing. At $d = 0$ the series is the raw price level; at $d = 1$ it is the ordinary return series, stationary and dominated by short-term fluctuation. The intermediate orders $d = 0.25$ and $d = 0.75$ produce series that retain progressively less of the slow-moving trend as d increases, tracing a continuous path between the two integer extremes.

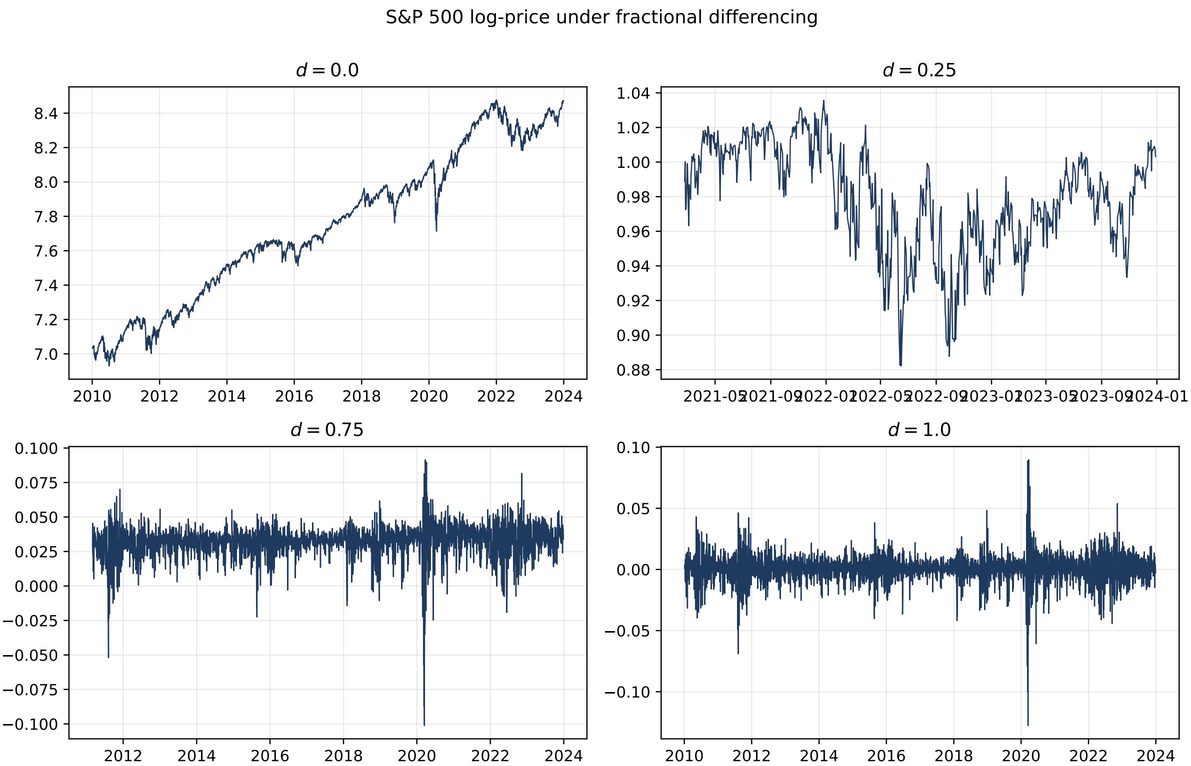

Figure 1: The S&P 500 price series under fractional differencing of orders $d = 0$ (the raw price level), $d = 0.25$, $d = 0.75$, and $d = 1$ (ordinary returns). As d increases from zero to one, the slow-moving trend is progressively removed and the series becomes increasingly dominated by short-term fluctuation.

2.2 Fourier Analysis and Frequency Responses

2.2.1 The Discrete Fourier Transform

Let $x_0, x_1, \ldots, x_{T-1}$ be a finite sequence of real-valued observations, for instance the daily returns of an asset over T trading days. The discrete Fourier transform re-expresses this sequence as a combination of complex exponentials, each oscillating at a distinct frequency. The transform is defined by

$$X_k = \sum_{t=0}^{T-1} x_t \, e^{-j2\pi kt/T}, \qquad k = 0, 1, \ldots, T-1, \tag{9}$$

where $j = \sqrt{-1}$ and each index k corresponds to the angular frequency $\omega_k = 2\pi k/T$ in radians per sample. The coefficient X_k is a complex number whose magnitude $|X_k|$

measures the amplitude of the oscillation at frequency ω_k and whose argument $\angle X_k$ measures its phase, that is, the position of the oscillation relative to the start of the sample. A large $|X_k|$ indicates that a cyclical component with period T/k samples accounts for a substantial part of the variation in the series.

The original sequence is recovered without loss through the inverse transform,

$$x_t = \frac{1}{T}\sum_{k=0}^{T-1} X_k\, e^{j2\pi kt/T}, \qquad t = 0, 1, \ldots, T-1. \tag{10}$$

Writing the complex exponential through Euler's identity, $e^{j\theta} = \cos\theta + j\sin\theta$, and using the conjugate symmetry $X_{T-k} = \overline{X_k}$ that holds because the input is real-valued, the inverse transform can be expressed entirely in terms of real trigonometric functions,

$$x_t = \frac{1}{T}\sum_{k=0}^{T-1} \Big(\Re[X_k]\cos\tfrac{2\pi kt}{T} - \Im[X_k]\sin\tfrac{2\pi kt}{T}\Big), \tag{11}$$

or equivalently, collecting the magnitude and phase of each coefficient, as

$$x_t = \frac{1}{T}\sum_{k=0}^{T-1} |X_k|\cos\Big(\tfrac{2\pi kt}{T} + \angle X_k\Big). \tag{12}$$

This last form makes the interpretation explicit: the series is a sum of cosine waves, one for each frequency ω_k, with amplitude $|X_k|/T$ and phase offset $\angle X_k$. The decomposition is exact rather than approximate, so no information is lost in moving between the time representation $\{x_t\}$ and the frequency representation $\{X_k\}$. The lowest non-zero frequency, $k = 1$, corresponds to a single cycle spanning the entire sample of T observations, while the highest, $k = \lfloor T/2 \rfloor$, corresponds to an oscillation that alternates at nearly every observation.

2.2.2 The Power Spectrum and the Decomposition of Variance

The Fourier coefficients also reveal how the variability of a series is distributed across timescales. The link is a classical result known as Parseval's theorem, which states that the total energy of a signal is the same whether measured in the time domain or the frequency domain. For a zero-mean series this energy is exactly the sample variance, and the theorem takes the form

$$\mathrm{var}(x_t) = \frac{1}{T}\sum_{t=0}^{T-1} x_t^2 = \frac{1}{T^2}\sum_{k=1}^{T-1} |X_k|^2. \tag{13}$$

The practical content of this identity is that the variance of the series can be split into separate, additive contributions, one from each frequency. Define the periodogram

$$\hat{S}_{xx}[k] = \frac{1}{T^2}|X_k|^2, \tag{14}$$

which is simply the share of the total variance carried by the oscillation at frequency ω_k. Summing these shares over all frequencies returns the variance in full; summing them over a subset of frequencies returns the part of the variance that lives in that band. The periodogram is therefore a budget: it tells the analyst exactly where the movement in a series comes from.

This is the heart of the matter, and it can be stated without any equation at all. Given a return series, one may ask what fraction of its total variation arises from rapid fluctuations between consecutive observations and what fraction arises from slow movements unfolding over many months. The power spectrum answers precisely this question. It takes the total variance, the single quantity into which ordinary risk measurement collapses all variation, and distributes it across a range of timescales, showing how much of an asset's variability is attributable to fast dynamics and how much to slow.

A few concrete shapes make the idea tangible. If a series were pure white noise, with no structure linking one observation to the next, its power would be spread evenly across all frequencies: the spectrum would be flat, and no timescale would carry more variance than any other. This is the benchmark against which real data is judged. A series dominated by slow trends, by contrast, concentrates its power at the low-frequency end: most of its variance comes from movements that take months or years to unfold, while the day-to-day variation is comparatively small. A series dominated by rapid reversals shows the opposite, with power piled up at the high-frequency end. The location of power along the frequency axis is thus a fingerprint of the dynamics of the series.

The translation from frequency to calendar time is direct and is what makes the spectrum economically meaningful. A frequency ω_k corresponds to a cycle that repeats every T/k observations, so for daily data a given point on the frequency axis maps to a definite number of trading days. Power concentrated at cycles of a few days reflects the rapid, largely transitory fluctuations associated with market microstructure and short-term reversals. Power at cycles of weeks to months reflects the medium-term dynamics associated with fund flows, rebalancing, and momentum. Power at cycles of a year or more reflects the slow components associated with business cycles, monetary regimes, and secular trends. When an analyst inspects the power spectrum of a return series, they are reading off how the asset's risk is apportioned among these horizons, an apportionment that the single summary number of variance conceals entirely.

This decomposition is the reason the frequency domain is the natural setting for the questions raised in Section 1. Two assets, or an asset and a benchmark, may carry the same total variance and yet distribute it across timescales in entirely different ways. An investor who cares about one horizon and not another is, in effect, interested in one region of the spectrum and indifferent to the rest. To make that preference precise, the variance must first be unpacked into its frequency components, which is exactly what the power spectrum provides. The next step is to extend the same decomposition from the variance of a single series to the covariance between two, since it is covariance with the benchmark that index tracking ultimately seeks to control.

2.2.3 The Cross-Spectrum and the Decomposition of Covariance

The same logic that splits the variance of one series across frequencies splits the covariance between two series in exactly the same way. This is the step that matters for index tracking, because tracking is fundamentally a statement about covariance: a portfolio replicates a benchmark when the two move together, and the quality of replication is governed by how their co-movement is distributed across timescales.

Let x_t and y_t be two zero-mean series, for instance the returns of an asset and the returns of a benchmark, with Fourier coefficients X_k and Y_k. The sample covariance decomposes as

$$\operatorname{cov}(x_t, y_t) = \frac{1}{T}\sum_{t=0}^{T-1} x_t y_t = \frac{1}{T^2}\sum_{k=1}^{T-1} \Re[X_k \overline{Y_k}]. \tag{15}$$

Each term in this sum is the contribution that frequency ω_k makes to the total covariance. Define the cross-periodogram

$$\hat{S}_{xy}[k] = \frac{1}{T^2}\Re[X_k \overline{Y_k}], \tag{16}$$

so that the covariance is the sum of these frequency-specific contributions, and the covariance restricted to a band of frequencies $\mathcal{K}$ is simply the sum taken over that band. Just as the periodogram is a budget for variance, the cross-periodogram is a budget for covariance: it shows at which timescales two series move together and at which they do not.

The crucial difference from the variance case is that the contributions can be negative. A frequency where the two series rise and fall together, in phase, contributes positively to the covariance. A frequency where one rises while the other falls, out of phase, contributes negatively. The total covariance is the net result of summing these signed contributions across all timescales. This is why two series can have low overall covariance for entirely different reasons: either they barely move together at any frequency, or they move together strongly at some frequencies and oppositely at others, with the contributions cancelling in the sum.

Consider an asset that tracks a benchmark closely on a day-to-day basis but slowly drifts away from it over the course of years, perhaps because it belongs to a sector in secular decline. Its high-frequency co-movement with the benchmark is strongly positive:

at the scale of days, the two are almost indistinguishable. Its low-frequency co-movement, however, is weak or even negative: at the scale of years, the asset and the benchmark part ways. The cross-periodogram of this pair would show large positive values at high frequencies and small or negative values at low frequencies. Now consider a second asset with the opposite profile: noisy and only loosely coupled to the benchmark from one day to the next, but firmly anchored to it over the long run, so that the two never wander far apart in level. This pair shows the reverse pattern, with weak high-frequency co-movement and strong positive low-frequency co-movement.

These two assets pose entirely different propositions to an index tracker, and which one is preferable depends entirely on the investor's horizon. The first asset is an excellent daily tracker and a poor long-term one; the second is a poor daily tracker and an excellent long-term one. An ordinary covariance calculation, which sums across all frequencies and reports a single number, cannot distinguish between them in this way: it might even assign them similar total covariances with the benchmark while concealing that the covariance is generated at opposite ends of the spectrum. The cross-spectrum makes the distinction explicit, and in doing so makes it possible to select assets according to the timescale at which co-movement with the benchmark is actually required.

This is the formal counterpart to the correlation and cointegration discussion of Section 1. The covariance of returns, on which the correlation paradigm is built, weights all frequencies of co-movement together, but because returns are differenced prices, that weighting is heavily tilted toward the high-frequency end, as the next section will make precise. Cointegration, by contrast, is a statement about co-movement at the very lowest frequencies, where the price levels of two integrated series remain tied together. The two paradigms are reading different entries in the same covariance budget. Once the budget is laid out frequency by frequency, the choice between them ceases to be a dichotomy and becomes a matter of selecting which entries to weight.

2.2.4 Linear Filters and the Reshaping of the Spectrum

The decompositions established in the previous two sections characterise the frequency content of a series as given. It remains to introduce the mechanism by which that content may be modified, so that specific timescales can be emphasised and others suppressed. This is the function of the linear filter.

A linear time-invariant filter transforms an input series x_t into an output series y_t through a weighted sum of past and present values,

$$y_t = \sum_{\tau} h_\tau \, x_{t-\tau}, \tag{17}$$

where the sequence $\{h_\tau\}$ is the impulse response that characterises the filter. In the time domain this operation, a convolution, combines neighbouring observations and its effect is not readily apparent. In the frequency domain it admits a simple description. The convolution theorem states that filtering multiplies each Fourier coefficient by a complex factor that depends only on the frequency,

$$Y_k = H(\omega_k)\, X_k, \qquad H(\omega) = \sum_{\tau} h_\tau\, e^{-j\omega\tau}, \tag{18}$$

where $H(\omega)$ is the frequency response of the filter. The defining property is that the filter does not transfer energy between frequencies; it scales each frequency independently. The oscillation at frequency ω_k in the output is the corresponding oscillation in the input, multiplied by the single factor $H(\omega_k)$.

The implication for the variance and covariance budgets is direct. Since each coefficient is scaled by $H(\omega_k)$, the periodogram of the output equals the periodogram of the input scaled by $|H(\omega_k)|^2$, and the cross-periodogram between two filtered series is scaled identically. Filtering therefore reweights the frequency-by-frequency contributions to variance and covariance, attenuating those timescales at which $|H(\omega)|$ is small and preserving those at which it approaches unity. The profile of $|H(\omega)|^2$ specifies exactly which timescales the filter retains and which it removes, so that the selection of a filter is equivalent to the selection of a weighting of the covariance budget.

Two filters anchor the analysis that follows, and they occupy opposite ends of the spectrum. The first is the moving average, which replaces each observation by the mean of the surrounding L values. Its frequency response is

$$H_{\mathrm{MA}}(\omega) = \frac{1}{L}\sum_{\tau=0}^{L-1} e^{-j\omega\tau}, \qquad |H_{\mathrm{MA}}(\omega)| = \frac{1}{L}\left|\frac{\sin(\omega L/2)}{\sin(\omega/2)}\right|, \tag{19}$$

which equals unity at zero frequency and decays toward zero as frequency increases, with the transition governed by the window length L. The moving average is accordingly a low-pass filter, passing slow movements and suppressing rapid ones, which formalises the sense in which smoothing removes short-term fluctuations. It follows that the covariance of two series each smoothed by a moving average equals the sum of the cross-spectrum over only the low frequencies the filter preserves. Smoothing the data prior to measuring covariance constitutes frequency-selective covariance estimation concentrated at long timescales.

The second filter is the first difference, which replaces each observation by its change from the preceding period, $y_t = x_t - x_{t-1}$. Its frequency response is

$$H_\Delta(\omega) = 1 - e^{-j\omega}, \qquad |H_\Delta(\omega)| = 2\left|\sin\tfrac{\omega}{2}\right|, \tag{20}$$

which vanishes at zero frequency and increases monotonically to its maximum at the highest frequency. Differencing is therefore a high-pass filter, removing slow movements and amplifying rapid ones, in direct contrast to the moving average. Measuring covariance on differenced data concentrates the estimate at short timescales. Figure 1 plots the magnitude responses of the two filters on a common axis, making the low-pass and high-pass characters and their mirrored structure apparent.

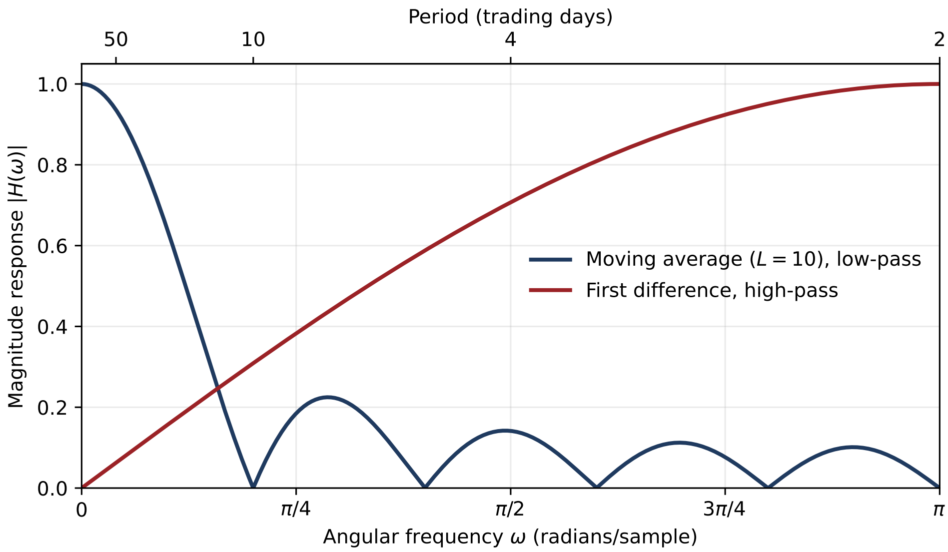

Figure 2: Magnitude responses of the moving-average (low-pass) and first-difference (high-pass) filters as functions of angular frequency. The moving average preserves low frequencies and attenuates high ones, while the first difference does the reverse; the two are mirror images across the frequency band.

These two filters are not arbitrary illustrations but the operations that distinguish the two paradigms of Section 1. Correlation-based tracking operates on returns, which are differenced prices, and therefore measures co-movement through a high-pass filter that emphasises the shortest timescales. Cointegration-based tracking operates on price levels, which are the cumulative sum of returns and hence the output of the extreme low-pass filter represented by integration, and therefore measures co-movement at the lowest frequencies. The two paradigms constitute the same covariance budget viewed through opposite filters. Section 3 derives these frequency responses in full, establishes the timescales to which they correspond under daily sampling, and demonstrates that the range of filters between them yields a continuum of tracking criteria, each matched to a distinct investor horizon.

2.3 Time-Frequency Representations

2.4 Index Tracking and Enhanced Index Tracking

2.5 Coherent Risk Measures and Linear Programme Formulations

2.6 Sparse Index Tracking

3 Frequency-Aware Index Tracking

3.1 Frequency Response of the Differencing Operator

3.2 Frequency Response of the Integration Operator

3.3 The Fractional Differencing Family

3.4 Passband Representations and Investor Horizon Targeting

3.5 Stationarity and the Need for Time-Frequency Representations

References

Artzner, P., Delbaen, F., Eber, J.-M., and Heath, D. (1999). Coherent measures of risk. *Mathematical Finance*, 9(3), 203–228.

Benidis, K., Feng, Y., and Palomar, D. P. (2018). Sparse portfolios for high-dimensional financial index tracking. *IEEE Transactions on Signal Processing*, 66(1), 155–170.

Alexander, C. (1999). Optimal hedging using cointegration. *Philosophical Transactions of the Royal Society of London, Series A*, 357(1758), 2039–2058.

Chaudhuri, S. E., and Lo, A. W. (2016). Spectral portfolio theory. *Working Paper*, MIT Laboratory for Financial Engineering.

Engle, R. F. (1974). Band spectrum regression. *International Economic Review*, 15(1), 1–11.

Euler, L. (1738). De progressionibus transcendentibus seu quarum termini generales algebraice dari nequeunt. *Commentarii Academiae Scientiarum Petropolitanae*, 5, 36–57.

Gençay, R., Selçuk, F., and Whitcher, B. (2002). *An Introduction to Wavelets and Other Filtering Methods in Finance and Economics*. Academic Press, San Diego.

Alexander, C., and Dimitriu, A. (2002). The cointegration alpha: Enhanced index tracking and long-short equity market neutral strategies. *ISMA Centre Discussion Papers in Finance*, 2002-08. University of Reading.

Alexander, C., and Dimitriu, A. (2005a). Indexing and statistical arbitrage: Tracking error or cointegration? *The Journal of Portfolio Management*, 31(2), 50–63.

Alexander, C., and Dimitriu, A. (2005b). Indexing, cointegration and equity market regimes. *International Journal of Finance and Economics*, 10(3), 213–231.

Alexander, C., Giblin, I., and Weddington, W. (2002). Cointegration and asset allocation: A new active hedge fund strategy. *Research in International Business and Finance*, 16, 65–90.

Arsov, S. (2023). Cointegration across stock markets in South-East Europe and possible spillovers. *SSRN Working Paper*. Available at SSRN 4640959.

Engle, R. F., and Granger, C. W. J. (1987). Co-integration and error correction: Representation, estimation, and testing. *Econometrica*, 55(2), 251–276.

Granger, C. W. J. (1981). Some properties of time series data and their use in econometric model specification. *Journal of Econometrics*, 16(1), 121–130.

Granger, C. W. J. (1988). Some recent developments on a concept of causality. *Journal of Econometrics*, 39(1–2), 199–211.

Granger, C. W. J., and Joyeux, R. (1980). An introduction to long memory time series models and fractional differencing. *Journal of Time Series Analysis*, 1(1), 15–29.

Grobys, K. (2010). Correlation versus cointegration: Do cointegration based index tracking portfolios perform better? Evidence from the Swedish stock market. *German Journal for Young Researchers*, 2(1), 72–78.

Hosking, J. R. M. (1981). Fractional differencing. *Biometrika*, 68(1), 165–176.

Johansen, S. (1988). Statistical analysis of cointegration vectors. *Journal of Economic Dynamics and Control*, 12(2–3), 231–254.

Johansen, S. (1991). Estimation and hypothesis testing of cointegration vectors in Gaussian vector autoregressive models. *Econometrica*, 59(6), 1551–1581.

Markowitz, H. (1959). *Portfolio Selection: Efficient Diversification of Investments*. John Wiley and Sons, New York.

Miao, G. J. (2014). High frequency and dynamic pairs trading based on statistical arbitrage using a two-stage correlation and cointegration approach. *International Journal of Economics and Finance*, 6(3), 96–110.

Roll, R. (1992). A mean/variance analysis of tracking error. *The Journal of Portfolio Management*, 18(4), 13–22.

Sant'Anna, L. R., Filomena, T. P., and Caldeira, J. F. (2016). Index tracking and enhanced indexing using cointegration and correlation with endogenous portfolio selection. *The Quarterly Review of Economics and Finance*, 65, 146–157.